

Leonard Thomas Aarons-Ditson

Quantum Information and Dynamics | Prospective Research in Near-Extremal Black-Hole Physics

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ACADEMIC PROFILE

Physicist with UCL postgraduate training and research activity in quantum information, quantum dynamics and many-body systems during postgraduate study, concluded in 2021. Formally trained in General Relativity, with subsequent experience using tensor notation, index methods and covariant formulations through graduate-level Quantum Field Theory and quantum theory. Now returning to academic research and extending these foundations into near-extremal black-hole physics, perturbation theory and QFT in curved spacetime.

EDUCATION

Postgraduate Certificate in Physics, University College London Distinction; awarded 1 Dec 2021

Postgraduate study undertaken within the MSc Physics programme from 2018 to 2021, including an interruption for cancer treatment; 90 UCL credits / 45 ECTS completed.

Selected postgraduate results: Advanced Quantum Theory 93.50; Modelling Quantum Many-Body-Systems 86.00; Quantum Computation and Communication 69.25; Quantum Field Theory 62.95.

Bachelor of Science in Physics, University College London 2013-2018

Second Class Honours (Lower Division). Formal undergraduate training included Mathematics for General Relativity. Selected results include Electricity and Magnetism 80; Mathematics for Physics and Astronomy 68; Quantum Physics 66; Mathematical Methods III 65; Quantum Mechanics 63; Theory of Dynamical Systems 61.

RESEARCH EXPERIENCE

Quantum-Information Research Activity, University College London Concluded 2021

During postgraduate study, worked within UCL's quantum-information community associated with Professor Sougato Bose. Investigated quantum communication through spin chains, entanglement and density-matrix evolution, including open and non-unitary dynamics and numerical approaches to time-dependent systems.

SELECTED AWARDS, PRESENTATIONS AND ACADEMIC ENGAGEMENT

Quantum Simulators, University College London During postgraduate study

Delivered a lecture on quantum simulation and quantum computation to an audience of students, alumni and academic staff.

"Quantum Computing: The End of Moore's Law?" During postgraduate study

Researched and wrote an article explaining quantum computation to non-specialists; commended for communicating a technically difficult subject clearly and concisely.

Cumberland Lodge Academic Selection, UCL Physics During BSc

Selected as one of 15 UCL Physics students to attend an academic programme at Cumberland Lodge, including a meeting with HM Queen Elizabeth II and HRH Prince Philip. Lectures from the professors and guest lecturers.

Recognition for Achievement in Physics and Mathematics, Shenfield High School 2013

At ages 16-17, independently teaching myself A-level Mathematics, Further Mathematics and Physics over approximately seven months and scoring A* A, with 100% in many of the 16 modules.

Talk at Imperial College on Emergence 2012

Delivered lecture to professors and classmates at the summer school event at imperial on Emergence phenomena age 16.

Imperial College London - Academic Initiative and Condensed-Matter Research Poster 2012

Initially approached Imperial independently at age 15 to ask about summer opportunities; later selected for a four-person project to interview researchers, synthesise their work and contribute to a condensed-matter research poster.

SELECTED RESEARCH AND COMPUTATIONAL PROJECTS

Investigating the infinite non-square potential

Collaborative numerical study of particle wavefunctions in non-standard potentials. Used Python, numerical methods, version control and a substantial IPython/Jupyter notebook to organise and present the work.

Quantum circuit simulations

Used Qiskit and custom gates to model spin-chain dynamics and time-dependent Hamiltonians, including exploratory work on Leggett-Garg inequalities.

SCIENTIFIC AND TECHNICAL SKILLS

Theoretical physics

Quantum information and computation; Quantum Dynamics; Spin Chains and Many-Body Systems; Density Matrices and Entanglement; Open and Non-Unitary Systems; Hilbert and Fock spaces; General Relativity; Tensor Calculus; Index notation; Covariant methods; graduate-level Quantum Field Theory.

Scientific computing

Python; NumPy; pandas; scikit-learn; Matplotlib; Jupyter; Qiskit; Git; numerical modelling; data analysis. Working knowledge of C++ and SQL.

PROFESSIONAL EXPERIENCE

999 Emergency Call handler, NHS - Chelmsford, Essex

Dec 2025-Jun 2026

Handled complex calls including suicide calls, cardiac arrest calls and managed complex medical high pressure situations.

Laboratory Analyst, NHS - Hammersmith Hospital, London

Mar 2022-Jan 2023

Performed high-throughput laboratory analysis supporting A&E and critical-care workflows, maintaining accuracy and clear prioritisation under strict turnaround requirements.

Data Scientist, Sticky, London

Oct 2020-Feb 2021

Used Python-based data analysis and machine-learning methods on product-placement and payment-data projects, and communicated technical results to colleagues from different backgrounds.

Data Analyst, Berry Palmer & Lyle (BPL Global), London

Jul-Sep 2017

Structured and migrated large document datasets and developed reusable knowledge-base assets to improve retrieval and reporting.

Project and data support, Canary Wharf Contractors, London

Jun-Oct 2016

Built simple analytical tools for workflow reporting and improved the organisation and accessibility of project records; received the company's Employee of the Year award in 2016.

ADDITIONAL ACTIVITIES

UCL Rowing

Sep 2015-Oct 2019

Competed for UCL; crew won Cambridge Head and placed second nationally in the novice category at Newcastle; trained more than 16-25 hours per week.